

CENTRE FOR ADVANCED STUDIES

**DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH,
LUCKNOW**



EVALUATION SCHEME AND SYLLABI

For

B.Tech. 2nd Year
Electronics Engineering (VLSI Design and Technology)

Effective from the Session: 2026-27

DR. A.P.J ABDUL KALAM TECHNICAL UNIVERSITY UTTAR PRADESH.
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DETAILED SYLLABI

B.Tech. Second Year

Effective-from Session 2026-27

B.Tech. (Electronics Engineering (VLSI Design and Technology)) Second Year, Semester-III

								Evaluation Scheme					
S N	Subject Code	Subject Name	Type	Category	Period			Sessional Component		Sessional (SW)(TS/PS)	End Semester Examination (ESE)	Total	Credit
					L	T	P	CT	TA	CT+T A	TE/PE	SW+ES E	Cr
1	CAS 302	Engineering Mathematics-III	T	ES	3	0	0	20	10	30	70	100	3
2	CEC 301	Network Analysis & Synthesis	T	ES	3	1	0	20	10	30	70	100	4
3	CEC 302	Electronics Devices & Circuits	T	ES	3	0	0	20	10	30	70	100	3
4	CEC 303	Digital Logic Design	T	ES	3	0	0	20	10	30	70	100	3
5	CEC 304	Signal & Systems	T	ES	3	1	0	20	10	30	70	100	4
6	CEC 351	Network Analysis and Synthesis Lab	P	ES	0	0	2	-	50	50	50	100	1
7	CEC 353	Digital Logic design Lab	P	ES	0	0	2	-	50	50	50	100	1
8	CEC 354	Electronics & Signal Systems Lab	P	ES	0	0	2	-	50	50	50	100	1
9	CVA 301	Constitution of India*	T	VA	2*	0	0	20	10	30*	70*	100*	1
					15+ 2*	2	6			300+ 30*	500+ 70*	800+100*	21
10.	Vocational Training / Skill based courses (Mandatory for Exit after II year)				-	-	-	-	-	-	-	-	2

*Compulsory Qualifying Audit Course

Abbreviation Used:

BS: Basic Science Course

ES: Engineering Science Course

VA: Value Added Course

B.Tech. (Electronics Engineering (VLSI Design and Technology)) Second Year, Semester-IV

										Evaluation Scheme			
SN	Subject Code	Subject Name	Type	Category	Period			Sessional Component		Sessional (SW)(TS/PS)	End Semester Examination (ESE)	Total	Credit
					L	T	P	CT	TA	CT+T A	TE/ PE	SW+ES E	Cr
1.	CEC 401	Microprocessor & Microcontroller	T	ES	3	0	0	20	10	30	70	100	3
2.	CEC 402	Control Systems	T	ES	3	1	0	20	10	30	70	100	4
3.	CEC 403	Analog Communication	T	ES	3	0	0	20	10	30	70	100	3
4.	CEC 404	Electromagnetic Field Theory & Waves	T	ES	3	0	0	20	10	30	70	100	3
5.	CEC 405	Semiconductor Fabrication Technology	T	ES	3	0	0	20	10	30	70	100	3
6.	CEC 451	Microprocessor Lab	P	ES	0	0	2	-	50	50	50	100	1
7.	CEC 452	Control Systems Lab	P	ES	0	0	2	-	50	50	50	100	1
8.	CEC 453	Communication Lab-I	P	ES	0	0	2	-	50	50	50	100	1
9.	CEC 456	Mini Project	P	ES	0	0	2	-	50	50	50	100	1
10	CVA 402	Universal Human Values*	T	VA	2*	0	0	20	10	30*	70*	100*	1
					15+ 2*	1	0 8			350+ 30*	550 + 70*	900+ 100*	21
11.	Vocational Training / Skill based courses (Mandatory for Exit after II year)				-	-	-	-	-	-	-	-	2

*Compulsory Qualifying Audit Course

Abbreviation Used:

BS: Basic Science Course

ES: Engineering Science Course

VA: Value Added Course

Summer Internship (4-week)/NPTEL Course (4 week) during summer break after Semester-II and same will be assessed / evaluated in the Semester-III

CAS 302: ENGINEERING MATHEMATICS-III

Course Objectives:

1. Remember the concept of partial differential equation and to solve partial differential equations.
2. Analyze the concept of partial differential equations to evaluate the problems concerned with partial differential equations
3. Understand the concept of correlation, moments, skewness and kurtosis and curve fitting.
4. Remember the concept of probability to evaluate probability distributions
5. Apply the concept of hypothesis testing and statistical quality control to create control charts

Unit	Topics	Lectures
I.	Partial Differential Equations :Origin of Partial Differential Equations, Linear and Non Linear Partial Equations of first order, Lagrange's Equations, Charpit's method, Cauchy's method of Characteristics, Solution of Linear Partial Differential Equation of Higher order with constant coefficients, Equations reducible to linear partial differential equations with constant coefficients.	8
II.	Applications of Partial Differential Equations : Classification of linear partial differential equation of second order, Method of separation of variables, Solution of wave and heat conduction equation up to two dimension, Laplace equation in two dimensions, Equations of Transmission lines.	8
III.	Statistical Techniques I : Introduction: Measures of central tendency, Moments, Moment generating function (MGF) , Skewness, Kurtosis, Curve Fitting , Method of least squares, Fitting of straight lines, Fitting of second degree parabola, Exponential curves ,Correlation and Rank correlation, Regression Analysis: Regression lines of y on x and x on y, regression coefficients, properties of regressions coefficients and non linear regression.	8
IV.	Statistical Techniques II: Probability and Distribution : Introduction, Addition and multiplication law of probability, Conditional probability, Baye's theorem, Random variables (Discrete and Continuous Random variable) Probability mass function and Probability density function, Expectation and variance, Discrete and Continuous Probability distribution: Binomial, Poission and Normal distributions.	8
V.	Statistical Techniques III: Sampling, Testing of Hypothesis and Statistical Quality Control : Introduction, Sampling Theory (Small and Large), Hypothesis, Null hypothesis, Alternative hypothesis, Testing a Hypothesis, Level of significance, Confidence limits.	8

Course Outcome (CO):		Bloom's Level
CO1	Solve linear and non-linear partial differential equations of first order and higher order using Lagrange's equations, Charpit's method, and Cauchy's method of characteristics.	K2, K5
CO2	Apply classification and separation of variables techniques to solve wave equations, heat conduction equations, and Laplace's equation, and analyze transmission line equations.	K1, K3
CO3	Apply measures of central tendency, moments, skewness, kurtosis and curve fitting techniques to analyze data, and determine correlation and regression relationships between variables.	K3
CO4	Apply the laws of probability and properties of random variables to determine probability mass/density functions and standard discrete and continuous probability distributions (Binomial, Poisson, Normal).	K1, K4
CO5	Apply sampling theory and hypothesis testing procedures to test statistical hypotheses and implement statistical quality control techniques.	K3, K4

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

Text Books

1. Erwin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability Theory, Universal Book Stall, 2003(Reprint).
3. S. Ross: A First Course in Probability, 6th Ed., Pearson Education India, 2002.
W. Feller, An Introduction to Probability Theory and its Applications, Vol. 1, 3rd Ed., Wiley, 1968

Reference Books:

- E. Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, 2005.
- Peter V. O'Neil, Advanced Engineering Mathematics, Thomson (Cengage) Learning, 2007.
- Maurice D. Weir, Joel Hass, Frank R. Giordano, Thomas, Calculus, Eleventh Edition, Pearson.
- D. Poole, Linear Algebra: A Modern Introduction, 2nd Edition, Brooks/Cole, 2005.
- Veerarajan T., Engineering Mathematics for first year, McGraw-Hill, New Delhi, 2008.
- Ray Wylie C and Louis C Barret, Advanced Engineering Mathematics, McGraw-Hill; Sixth Edition.
- P. Sivaramakrishna Das and C. Vijayakumari, Engineering Mathematics, 1st Edition, Pearson Education.
- Advanced Engineering Mathematics. Chandrika Prasad, Reena Garg, 2018.
- H.K. Dass, Advanced Engineering Mathematics, H.K. Dass, S. Chand

CEC 301: NETWORK ANALYSIS AND SYNTHESIS

Course Objectives:

1. Apply basics of electrical circuits with nodal and mesh analysis.
2. Illustrate electrical network theorems.
3. Understand the concept of Laplace and Fourier transform and transform circuits using Thevenin's and Norton's theorem.
4. Analyze electrical circuits under transient and steady state conditions.

Unit	Topics	Lectures
I.	Node and mesh analysis, matrix approach of network containing voltage & current sources and reactances, source transformation and duality	8
II.	Network theorems: Superposition, reciprocity, Thevenin's, Norton's, Maximum power transfer, compensation and Tellegen's theorem as applied to A.C. circuits.	8
III.	Trigonometric and exponential Fourier series: Discrete spectra and symmetry of waveform, steady state response of a network to non-sinusoidal periodic inputs, power factor, effective values, Fourier transform and continuous spectra	8
IV.	Laplace transforms and properties: Partial fractions, singularity functions, wave form synthesis, analysis of RC, RL, and RLC networks with and without initial conditions with Laplace transforms evaluation of initial conditions.	8
V.	AI in circuit analysis: AI-based equation solving, Automatic circuit simplification AI in Network Theorems :AI-based Thevenin/Norton equivalent calculation and optimization of circuit parameters. AI-based Simulation: Circuit simulation using: LT spice AI-assisted waveform prediction	8

Course Outcomes:

Course Outcome (CO)		Bloom's Level
At the end of this course, the students will be able to:		
CO 1	Analyze electrical networks containing voltage and current sources using node and mesh analysis, matrix methods, source transformation and duality.	K2, K4
CO 2	Apply network theorems (Superposition, Reciprocity, Thevenin's, Norton's, Maximum Power Transfer, Compensation, Tellegen's) to analyze A.C. circuits.	K1& K3
CO 3	Analyze the response of networks to non-sinusoidal periodic inputs using trigonometric and exponential Fourier series, and evaluate power factor and effective values.	K3
CO 4	Apply Laplace transform techniques, partial fractions and singularity functions to analyze RC, RL and RLC networks with and without initial conditions.	K1, K4
CO 5	Apply AI-based tools and techniques for circuit equation solving,	K3, K4

	automatic circuit simplification, Thevenin/Norton equivalent calculation, and circuit simulation.	
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K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

Text Book

1. A. Anand Kumar, “Network Analysis and Synthesis”, PHI publication, 2019.
2. A.K Chakraborty, “Network Analysis and Synthesis”, Mc Graw Hill publication, 2018.

Reference Books

- Franklin F. Kuo, “Network Analysis and Synthesis”, Wiley India Education, 2nd Ed., 2006.
- Van, Valkenburg, “Network analysis”, Pearson, 2019.
- Sudhakar, A., Shyammoan, S.P., “Circuits and Network, “Tata McGraw-Hill New Delhi, 1994.
- A William Hayt, “Engineering Circuit Analysis”, 8th Edition, McGraw-Hill Education

CEC 302: ELECTRONICS DEVICES AND CIRCUITS

Course Objectives:

1. Get familiar with quantum mechanics, semiconductor physics and energy bands.
2. Knowledge about excess carrier in semiconductor material and its various Junction Properties.
3. Analyze BJT as amplifier, its/high frequency response and small signal models
4. To understand working of MOS capacitor, MOSFET and their characteristics
5. To understand the concept of feedback topologies and various oscillators.

UNIT	Topic	Lectures
1	Various classes of operation (Class A, B, AB, C etc.), their power efficiency and linearity issues, Oscillators: Review of the basic concept, Barkhausen criterion, RC oscillators (phase shift, Wien bridge etc.), LC oscillators (Hartley, Colpitt, Clapp etc.), non sinusoidal oscillators.	8
2	BJT amplifier models: Voltage amplifier, current amplifier, trans-conductance amplifier and trans-resistance amplifier. Biasing schemes for BJT and FET amplifiers, bias stability, various configurations (such as CE/CS, CB/CG, CC/CD) and their features, small signal analysis, low frequency transistor models, estimation of voltage gain, input resistance, output resistance etc.,	8
3	High frequency transistor models, design procedure for particular specifications, low frequency analysis of multistage amplifiers MOSFET: Review of device structure operation and V-I characteristics. Circuits at DC, MOSFET as Amplifier and switch, Biasing in MOS amplifier circuits, small-signal operation and models, single stage MOS amplifier, MOSFET internal capacitances and high frequency model, frequency response of CS amplifier	8
4	Feedback: The general feedback structure, properties of negative feedback, the four basic feedback topologies: Voltage series, current series, voltage shunt, current shunt, effect of feedback on gain, bandwidth etc., calculation with practical circuits, concept of stability, gain margin and phase margin.	8
5	Introduction to AI in Electronics Artificial Intelligence Introduction (Basics, Types: ML, DL) Significance of AI in Electronics, Traditional vs AI-based circuit analysis Data-driven modeling of electronic devices AI tools for circuit design (like MATLAB, Python libraries)	8

Course Outcomes:

CO	Course Outcome	Bloom's Level
CO1	Comprehensive understanding of semiconductor physics, carrier transport, P-N junctions, I-V characteristics, and device modeling.	K1,K3
CO2	Calculate and analyze carrier lifetime, understanding its impact on device performance and operation.	K2
CO3	Understand the concept of BJT operation, amplification, biasing, small-signal models, and frequency response in amplifier circuits.	K3,K4
CO4	Analyze the concept of Biasing in MOS amplifier circuits and high	K4

	frequency models.	
CO5	Understand the concept of feedback topologies and principles of oscillator circuits.	K4

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

Text Books:

1. B.G. Streetman and S.Banerjee “Solid state electronics devices”, 6th Edition, PHI.
2. A.S. Sedra and K.C. Smith, “Microelectronic Circuits”, Oxford University Press, 5th Ed

Reference Books:

- D.Neamen, D.Biswas", Semiconductor Physics and Devices", McGraw-Hill Education.
- J.Millman and A.Grabel, “Microelectronics”, 2nd edition, McGraw Hill, 1988.
- Paul R.Gray and Robert G.Meyer, “Analysis and Design of Analog Integrated Circuits,” John Wiley, 3rd edition.
- Muhammad H. Rashid, “Electronic Devices and Circuits”, Cengage publication, 2014.

CEC 303: DIGITAL LOGIC DESIGN

Pre-requisite: Fundamentals of Electronics Engineering

Course Objectives:

1. Introduce the concept of digital and Binary system.
2. Design and analyze combination logic circuits.
3. Design and analyze sequential logic circuits.
4. Design and analyze DAC and ADC circuits, identifying the strengths and limitations of different architectures
5. Use Hardware Description Languages (HDLs) to model and simulate digital designs, and understand the design flow from HDL to hardware implementation.

Unit	Topics	Lect ures
1	Intelligent Combinational circuit logic & design: MSI device like Half and full adders, subtractors, serial and parallel adder, BCD adder, Decoders, Encoders, Multiplexer, Demultiplexers & Magnitude comparator. AI-based design of: Adders, Subtractors, Multiplexers / Demultiplexers, Encoders/Decoders Automated circuit generation using AI Truth table using circuit synthesis using ML	8
2	Sequential Logic Circuits Design: Introduction to Sequential logic design: Building blocks like S-R, JK and Master-Slave JK FF, edge triggered FF, characteristics table. Analysis of Clock Sequential circuits: state table, Reduction of state table, Excitation table State Diagram design of clock sequential circuits. Synchronous and asynchronous counters, shift registers, Introduction to finite state machines, design of synchronous FSM, algorithmic state machines charts. Designing synchronous circuits like pulse train generator, pseudo random binary Sequence generator, clock generation.	8
3	Logic families and semiconductor memories: Introduction to Logic Families, characteristics Logic Families, fan-in, fan-out, Power Dissipation Speed of operation, noise margin, Operating Temperature, Tristate TTL and NAND gate, specifications, propagation delay, tristate TTL, ECL, CMOS families and their interfacing, MUX implementation using CMOS, memory elements, concept of programmable logic devices like PAL, PLA and FPGA, logic implementation using programmable devices.	8
4	Digital-to-Analog converters (DAC): Introductions to Digital-to-Analog converters using Weighted resistor, R-2R ladder and resistor string etc. Analog-To- Digital converters (ADC): Using single slope, dual slope, successive approximation, and flash etc. switched capacitor circuits: Basic concept, practical configurations, application in amplifier, integrator, ADC etc.	8

5	Introduction to AI based Hardware Description Languages (HDL) code generation: Introduction to HDL, Design flow using HDL (VHDL or Verilog), Behavioral, Dataflow, and Structural modeling, Simulation and verification of digital designs, Introduction to synthesis and implementation tools, Implementation of combinational circuits Adder, subtractor and MUX, Implementation of sequential circuits: Latch, flip flops and counters.	8
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Course Outcomes:

CO	Course Outcome	Bloom's Level
CO1	Design and analyze combinational logic circuits such as adders, subtractors, decoders, encoders, multiplexers and demultiplexers, including AI-based circuit synthesis techniques.	K1,K3
CO2	Design and analyze sequential logic circuits including flip-flops, counters, shift registers and finite state machines.	K2,K4
CO3	Describe the characteristics of various logic families and semiconductor memory/programmable logic devices such as PAL, PLA and FPGA.	K3,K1
CO4	Design and analyze Digital-to-Analog and Analog-to-Digital converter circuits and switched capacitor circuits.	K4,K5
CO5	Use Hardware Description Languages (VHDL/Verilog) to model, simulate and implement combinational and sequential digital circuits.	K5,K3

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

Text Books:

1. M.Morris Mano and M.D. Ciletti, “Digital Design”, Pearson Education.
2. S.Salivahanan and S.Arivazhagan, “Digital Circuits and Design” 3rd edition, 2009.

Reference Books:

- R.P.Jain, “Modern Digital Electronics”, Tata Mc Graw Hill, 4th edition, 2009.
- Anand Kumar, “Fundamental of Digital Circuits”. PHI 4th edition, 2018.
- W.H. Gothmann, “Digital Electronics-An Introduction to Theory and Practice”, PHI, 2nd edition, 2006.
- D.V.Hall, “Digital Circuits and Systems”, Tata Mc Graw Hill.

CEC 304: SIGNAL & SYSTEMS

Course Objectives:

1. To understand the behavior of signals in time and frequency domain
2. To understand the characteristics of LTI systems
3. To give the concepts of signals and systems and its analysis using different transform techniques
4. To understand the Sampling and Reconstruction of signal

Unit	Topics	Lectures
I.	SIGNALS AND THEIR REPRESENTATION Analogy between vectors and signals, Orthogonal signal space, Orthogonality in complex functions, Eigen analysis, Continuous-time and Discrete-time signals, Energy and Power signals, Periodic and Aperiodic signals, Even and Odd signals, Exponential and sinusoidal signals etc., Transformations of the independent variable, Concepts of Unit impulse and Unit sample signals, Signum function. Continuous-time and Discrete-time systems and basic system properties.	8
II.	LINEAR TIME-INVARIANT (LTI) SYSTEMS AND FOURIER SERIES REPRESENTATION OF PERIODIC SIGNALS Discrete and Continuous time LTI systems, Convolution Sum, Convolution Integral, Properties of LTI systems, Causal LTI systems described by difference equations, Singularity functions. Continuous-time and Discrete time signals and their Fourier Series representation, Properties of Fourier Series, Dirichlet's Conditions, Complex Fourier Spectrum.	8
III.	REPRESENTATION OF APERIODIC SIGNALS BY FOURIER TRANSFORMS AND LAPLACE TRANSFORMS Continuous-time and Discrete-time signals and their Fourier Transforms, Fourier Transforms of periodic signals and standard signals, Properties of Fourier Transforms, System characterized by linear constant coefficient differential equation. Principles of Laplace Transform, The Region of Convergence (ROC), Properties of ROC, Relationship between Laplace Transform and Fourier Transform, Properties of Laplace transform.	8
IV.	Z-TRANSFORMS Principles of Z-Transform, The Region of Convergence (ROC), Properties of ROC, Relationship between Z-Transform and Fourier Transform, Properties of Z-transform, Inverse Z-transform, Pole zero plot, Power series expansion and Partial Fraction Expansion, Initial value and Final value Theorems, Analysis and characterization of LTI system using Z-Transforms.	8
V.	AI Applications in Signals & Systems: Speech signal processing (voice assistants), Image processing basics, Biomedical signal analysis (ECG), Sensor data processing (IoT systems), Pattern recognition basics.	8

Course Outcomes:

CO	Course Outcome	Bloom's Level
CO1	Understand the mathematical description and representation of continuous and discrete time signals and systems.	K1,K2
CO2	Develop input output relationship for linear time shift invariant systems and understand the convolution operator for continuous and discrete time systems.	K2,K3
CO3	Understand and resolve the signals in frequency domain using Fourier series and Fourier transforms.	K1,K3
CO4	Understand the limitation of Fourier transform and need for Laplace transform and Z transform and develop the ability to analyse the system in s domain and z domain.	K2,K4
CO5	Understand the sampling theorem, aliasing, types of sampling, and reconstruction of signal from its samples along with the introduction of random signals and probability theory.	K1,K2

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

Text Books:

1. “Signals and Systems” by A.V. Oppenheim, A.S Willsky, and S.H. Nawab, Prentice-Hall, Englewood Clieffs.

Reference Books:

- “Fundamentals of Signals and Systems” by Michel J. Robert, MGH International Edition
- “Signals & Systems & Communication” by B.P.Lathi, Bsp
- “Signals and Systems” by Simon Haykin and Van Veen, Wiley, 2nd Edition.
- “Probability, random variables and stochastic Processes” by A Papoulis, McGraw-Hill.

CEC 351: NETWORK ANALYSIS AND SYNTHESIS LAB

Course Objectives:

1. Understand and verify the fundamental concepts of electrical network analysis through experimental investigations and simulations.
2. Analyze the behavior of linear and nonlinear electrical networks using circuit laws, network theorems, and mathematical techniques.
3. Verify network theorems such as Superposition, Thevenin's, Norton's, Maximum Power Transfer, Reciprocity, and Compensation through practical experiments.
4. Study transient and steady-state responses of RL, RC, and RLC circuits and interpret their significance in electrical and electronic systems.
5. Analyze two-port networks and determine network parameters (Z , Y , h , and $ABCD$) using experimental and simulation-based approaches.

SUGGESTIVE LIST OF EXPERIMENTS

1. Verification of Kirchhoff's laws.
2. Verification of Superposition theorem.
3. Verification of Thevenin's Theorem and Maximum power transfer theorem.
4. Verification of Tellegen's theorem.
5. Measurement of power and power factor in a single phase AC series inductive circuit and study improvement of power factor using capacitor.
6. Study of phenomenon of resonance in RLC series circuit and obtain resonant frequency.
7. Determination of parameters of AC single phase series RLC circuit.
8. To find poles and zeros of immittance function.
9. Design and find cut-off frequency of low pass and high pass filters.
10. Design and find the pass band frequencies of band pass filters.
11. Design and find the stop band frequencies of band reject filters.

Course Outcome (CO): At the end of this course students will demonstrate the ability to		Bloom's Level
CO1	Understand basics of electrical circuits with nodal and mesh analysis.	K2
CO2	Appreciate electrical network theorems.	K2
CO3	Analyze RLC circuits.	K3
CO4	Determine the stability of an electrical circuit.	K4
CO5	Design network filters.	K4

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate

Related Lab Video Links:

<https://www.youtube.com/@electronicsplan t3272>

<https://www.youtube.com/watch?v=6-2AQWzjxuY&t=43s>

<https://www.youtube.com/watch?v=fBpCY12jxPU&t=33s>

<https://www.youtube.com/watch?v=ruuGNSpVrMw><https://youtu.be/Hv06P-RlQSU>

CEC 353: DIGITAL LOGIC DESIGN LAB

Course Objectives:

1. Understand and verify the operation of basic logic gates, Boolean algebra, and digital logic principles through practical implementation.
2. Design, analyze, and implement combinational logic circuits such as adders, subtractors, multiplexers, demultiplexers, encoders, decoders, and comparators.
3. Develop and test sequential logic circuits including flip-flops, registers, counters, and shift registers using digital ICs and simulation tools.
4. Apply logic minimization techniques such as Karnaugh Maps (K-Maps) for efficient digital circuit design.
5. Analyze timing characteristics and functional behavior of digital systems through hardware implementation and simulation.

SUGGESTIVE LIST OF EXPERIMENTS

1. Introduction to digital electronics lab- nomenclature of digital ICs, specifications, study of the datasheet, Concept of Vcc and ground, verification of the truth tables of logic gates using TTL ICs.
2. Implementation of the given Boolean function using logic gates in both SOP and POS forms.
3. Verification of state tables of RS, JK, T and D flip-flops using NAND & NOR gates.
4. Implementation and verification of Decoder using logic gates.
5. Implementation and verification of Encoder using logic gates.
6. Implementation of 4:1 multiplexer using logic gates.
7. Implementation of 1:4 demultiplexer using logic gates.
8. Implementation of 4-bit parallel adder using 7483 IC.
9. Design, and verify the 4-bit synchronous counter.
10. Design, and verify the 4-bit asynchronous counter.

CEC 354: ELECTRONICS AND SIGNAL SYSTEM LAB

Course Objectives:

1. Understand the frequency response and gain characteristics of single-stage and multistage transistor amplifiers.
2. Analyze the effect of various feedback topologies on amplifier performance, including gain, bandwidth, stability, and input/output impedances.
3. Investigate the characteristics and performance of FET and MOSFET amplifiers by measuring gain, bandwidth, input impedance, and maximum signal handling capacity.
4. Measure important operational amplifier parameters such as differential gain, common-mode gain, CMRR, and slew rate using standard laboratory techniques.
5. Develop programming skills for generating and analyzing discrete-time signals and digital systems using MATLAB/Octave or equivalent software.

Implement following programs on MATLAB, Calibre software.

SUGGESTIVE LIST OF EXPERIMENTS

1. Study of Multi-stage amplifiers: Frequency response of single stage and multistage amplifiers.
2. Feedback topologies: Study of voltage series, current series, voltage shunt, current shunt, effect of feedback on gain, bandwidth etc.
3. Field effect transistors: Single stage common source FET amplifier –plot of gain in dB vs frequency, measurement of bandwidth, input impedance, maximum signal handling capacity (MSHC) of an amplifier.
4. Metal Oxide Semiconductor Field Effect Transistors: Single stage MOSFET amplifier plot of gain in dB Vs frequency, measurement of, bandwidth, input impedance, maximum signal handling capacity (MSHC) of an amplifier.
5. Measurement of Op-Amp parameters: Common mode gain, differential mode gain, CMRR, slew rate.
6. Write a program to generate the following discrete sequences:-
(i) Unit Step (ii) Unit Impulse (iii) Unit Ramp (iv) Periodic sinusoidal sequences. Plot all the sequences.
7. Write a program to find the magnitude and phase response of the first order low pass and high pass filter. Plot the responses in logarithmic scale.
8. Write a program to find the response of a low pass filter and high pass filter, when a speech signal is passed through these filters.
9. Write a program to convolve two discrete time sequences and plot all the sequences. Verify the result by analytical calculation.
10. Find the Fourier transform of a square pulse. Plot its amplitude and phase spectrum.

Course Outcome (CO): At the end of this course students will demonstrate the ability to		Bloom's Level
CO1	Analyze the frequency response of single-stage and multistage amplifiers and determine important parameters such as gain and bandwidth.	K2
CO2	Evaluate different feedback amplifier configurations and explain the effect of feedback on gain, bandwidth, stability, and amplifier performance.	K2
CO3	Design and experimentally verify the characteristics of FET and MOSFET amplifiers by measuring gain, bandwidth, input impedance, and maximum signal handling capacity.	K3
CO4	Measure and interpret important operational amplifier parameters	K4

	including differential gain, common-mode gain, CMRR, and slew rate using laboratory instruments.	
CO5	Develop MATLAB/Octave programs to generate discrete-time signals, perform convolution, analyze digital filters, and evaluate system responses.	K4

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate

CVA-301 CONSTITUTION OF INDIA (MANDATORY — 1 CREDIT)

Course Objectives

The objective of this course is to familiarize students with the fundamental principles of the Indian Constitution and its significance in democratic governance.

The students will learn:

- Understand the historical background, making and salient features of the Indian Constitution.
- Study Fundamental Rights, Fundamental Duties and Directive Principles of State Policy.
- Understand the structure and functioning of the Parliamentary and Federal systems of India.
- Study Constitutional amendments, Emergency provisions and Local Self Government schemes.
- Appreciate the role of the Constitution in shaping Indian democracy and governance.

Syllabus

Unit 1: Introduction and Basic Information about Indian Constitution	Contact Hours
Meaning of the constitution law and constitutionalism, Historical Background of the Constituent Assembly, Government of India Act of 1935 and Indian Independence Act of 1947, Enforcement of the Constitution, Indian Constitution and its Salient Features, The Preamble of the Constitution, Fundamental Rights, Fundamental Duties, Directive Principles of State Policy, Parliamentary System, Federal System, Centre-State Relations, Amendment of the Constitutional Powers and Procedure, The historical perspectives of the constitutional amendments in India, Emergency Provisions: National Emergency, President Rule, Financial Emergency, and Local Self Government – Constitutional Scheme in India.	8
Unit 2: Union Executive and State Executive	
Powers of Indian Parliament Functions of Rajya Sabha, Functions of Lok Sabha, Powers and Functions of the President, Comparison of powers of Indian President with the United States, Powers and Functions of the Prime Minister, Judiciary – The Independence of the Supreme Court, Appointment of Judges, Judicial Review, Public Interest Litigation, Judicial Activism, Lokpal, Lok-Ayukta, The Lokpal and Lok-Ayuktas Act 2013, State Executives – Powers and Functions of the Governor, Powers and Functions of the Chief Minister, Functions of State Cabinet, Functions of State Legislature, Functions of High Court and Subordinate Courts.	8
Unit 3: Introduction and Basic Information about Legal System	
The Legal System: Sources of Law and the Court Structure: Enacted law -Acts of Parliament are of primary legislation, Common Law or Case law, Principles taken from decisions of judges constitute binding legal rules. The Court System in India and Foreign Courtiers (District Court, District Consumer Forum, Tribunals, High Courts, Supreme Court). Arbitration: As an alternative to resolving disputes in the normal courts, parties who are in dispute can agree that this will instead be referred to arbitration. Contract law, Tort, Law at workplace.	8

Unit 4: Intellectual Property Laws and Regulation to Information	
Intellectual Property Laws- Introduction, Legal Aspects of Patents, Filing of Patent Applications, Rights from Patents, Infringement of Patents, Copyright and its Ownership, Infringement of Copyright, Civil Remedies for Infringement, Regulation to Information-Introduction, Right to Information Act, 2005, Information Technology Act, 2000, Electronic Governance, Secure Electronic Records and Digital Signatures, Digital Signature Certificates, Cyber Regulations Appellate Tribunal, Offences, Limitations of the Information Technology Act.	8
Unit 5: Business Organizations and E-Governance	
Sole Traders, Partnerships: Companies: The Company's Act: Introduction, Formation of a Company, Memorandum of Association, Articles of Association, Prospectus, Shares, Directors, General Meetings and Proceedings, Auditor, Winding up. E-Governance and role of engineers in E-Governance, Need for reformed engineering serving at the Union and State level, Role of I.T. professionals in Judiciary.	8

Course Outcomes:

CO	Course Outcome (CO)	Bloom's Level
CO1	Understand the historical background, making, and salient features of the Indian Constitution including Fundamental Rights and Duties.	K2
CO2	Analyze the structure and functions of the Union and State Executives, and the role of the Judiciary in Indian governance.	K4
CO3	Understand the legal system, sources of law, court structure, and alternative dispute resolution mechanisms.	K2
CO4	Understand intellectual property laws and regulations governing information technology and digital governance	K2
CO5	Apply knowledge of business organizations and e-governance to understand the role of engineers in democratic society.	K4

K1 – Remember, K2 – Understand, K3 – Apply, K4 – Analyze, K5 – Evaluate

Text Books

1. Brij Kishore Sharma; Introduction to the Indian Constitution, PHI, New Delhi, latest edition.
2. Granville Austin; The Indian Constitution: Corner stone of a Nation. 1966, Oxford Clarendon Press.
3. Subhash C. Kashyap; Our Constitution: An Introduction to India's Constitution and 35 constitutional Law, NBT, 2018.
4. P. M. Bakshi; The Constitution of India, Latest Edition, Universal Law Publishing.

Reference Books:

1. V. K. Ahuja; Law Relating to Intellectual Property Rights (2007)
2. Suresh T. Viswanathan: The Indian Cyber Laws, Bharat Law House, New Delhi-88
3. P. Narayan; Intellectual Property Law, Eastern Law House, New Delhi

Syllabus

B.Tech 2nd Year

IV Semester

CEC-401: MICROPROCESSOR & MICROCONTROLLER

Course Objective:

1. Understand the architecture, organization, and operation of microprocessors and microcontrollers, including their functional units and instruction sets.
2. Analyze and develop assembly language programs for arithmetic, logical, data transfer, and control operations using microprocessor and microcontroller architectures.
3. Understand memory organization, addressing modes, interrupts, and interfacing techniques used in microprocessor-based systems.
4. Design and implement interfacing of peripheral devices such as keyboards, displays, ADCs, DACs, timers, and communication modules with microprocessors and microcontrollers.
5. Apply microcontroller-based solutions for embedded system applications involving sensing, control, automation, and real-time processing.

Unit	Topics	Lectures
I	Assembly programming and instruction of 8051: Introduction to 8051 assembly programming, Assembling and running an 8051 program, Data types and Assembler directives, Arithmetic, logic instructions and programs, Jump, loop and call instructions, IO port programming. Programming 8051 Timers. Serial Port Programming, Interrupts Programming, Interfacing: LCD & Keyboard Interfacing, ADC, DAC & Sensor Interfacing, External Memory Interface, Stepper Motor and Waveform generation.	8
II	Basic Programming concepts: , Flow chart symbols, Data Transfer operations, Arithmetic operations, Logic Operations, Branch operation, Writing assembly language programs, Programming techniques: looping, counting and indexing. Additional data transfer and 16 bit arithmetic instruction, Logic operation: rotate, compare, counter and time delays, 8085 Interrupts.	8
III	AI-supported Microprocessor Architecture: 16-bit Microprocessors (8086): Architecture, Pin Description, Physical address, segmentation, memory organization, Addressing modes. Peripheral Devices: 8237 DMA Controller, 8255 programmable peripheral interface, 8253/8254 programmable timer/counter, 8259 programmable interrupt controller, 8251 USART and RS232C.	8
IV	8051 Microcontroller Basics: Inside the Computer, Microcontrollers and Embedded Processors, Block Diagram of 8051, PSW and Flag Bits, 8051 Register Banks and Stack, Internal Memory Organization of 8051, IO Port Usage in 8051, Types of Special Function Registers and their uses in 8051, Pins Of 8051. Memory Address Decoding, 8031/51 Interfacing With External ROM and RAM.8051 Addressing Modes.	8
V	Introduction to AI in Embedded Systems: Overview of Artificial Intelligence, AI vs Embedded Systems, Microprocessor vs Microcontroller with AI perspective, Edge AI, AI-enabled embedded system architecture Microprocessor architecture and its operations, Memory, Input & output devices, The 8085 MPU- architecture, Pins and signals, Timing	

	Diagrams, Logic devices for interfacing, Memory interfacing, Interfacing output displays, Interfacing input devices, Memory mapped I/O.	
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Course Outcome (CO)		Bloom's Level
CO1	Demonstrate the basic architecture of 8085.	K1
CO2	Illustrate the programming model of microprocessors & write program using 8085 microprocessors. AI model optimization (quantization, pruning)	K2
CO3	Demonstrate the basics of 8086 Microprocessor and interface different external Peripheral Devices like timer, USART etc. with Microprocessor (8085/8086).	K2, K3
CO4	Compare Microprocessors & Microcontrollers, and comprehend the architecture of 8051 microcontroller	K2, K4
CO5	Illustrate the programming model of 8051 and implement them to design projects on real time problems.	K3, K4

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate

Text Books:

1. Ramesh Gaonkar, "Microprocessor Architecture, Programming, and Applications with the 8085", 6th Edition, Penram International Publication (India) Pvt. Ltd.,2013
2. D.V.Hall:Microprocessors Interfacing,TMH 3rd Edition,
3. Mazidi Ali Muhammad, Mazidi Gillispie Janice, and Mc Kinlay Rolin D., "The 8051 Microcontroller and Embedded Systems using Assembly and C", Pearson, 2nd Edition,2006

Reference Books:

- Kenneth L. Short, "Microprocessors and programmed Logic", 2nd Ed, Pearson Education Inc.,2003
- Barry B. Brey, "The Intel Microprocessors, 8086/8088, 80186/80188, 80286, 80386, 80486, Pentium, Pentium Pro Processor, Pentium II, Pentium III, Pentium IV, Architecture, Programming & Interfacing", Eighth Edition, Pearson Prentice Hall, 2009.
- Shah Satish, "8051 Microcontrollers MCS51 Family and its variants", Oxford, 2010

CEC – 402: CONTROL SYSTEM

Course Objectives:

1. Understand the fundamental concepts, principles, and classifications of control systems, including open-loop and closed-loop systems.
2. Develop mathematical models of physical systems using differential equations, transfer functions, block diagrams, and signal flow graphs.
3. Analyze the time-domain and frequency-domain performance of control systems and evaluate system characteristics such as stability, accuracy, and transient response.
4. Apply various stability analysis techniques, including Routh–Hurwitz Criterion, Root Locus, Nyquist Criterion, and Bode Plot methods.
5. Design and evaluate controllers and compensators to improve system performance and meet specified design requirements.

Unit	Topics	Lectures
I	Introduction to Control Systems & AI: basic overview of Control Systems (Open loop / Closed loop), Classical vs Modern Control Systems, introduction of Artificial Intelligence , AI in control systems: Importance and scope, Limitations of conventional controllers, AI-based control architecture	8
II	Intelligent Control Techniques: Introduction to Intelligent Control , ANN Overview, Rule-based systems, Expert systems in control, Fuzzy Logic Control (FLC), Neural Network based controllers, Adaptive and self-tuning controllers State-Variable Analysis: Introduction, vector matrix representation of state equation, state transition matrix, state-transition equation, relationship between state equations and high- order differential equations, relationship between state equations and transfer functions, Decomposition of transfer functions, Controllability and observability, Eigen Value and Eigen Vector, Diagonalization.	8
III	Time domain Analysis of Control Systems: Time response of continuous data systems, typical test signals for the time response of control systems, unit step response and time- domain specifications, time response of a first order system	8
IV	Stability of Linear Control Systems: Bounded-input bounded-output stability continuous data systems, zero-input and asymptotic stability of continuous data systems, RouthHurwitz criterion, Root-Locus Technique: Introduction, Properties of the Root Loci, Design aspects of the Root Loci.	8
V	Neural Network Basics for Control: Artificial Neural Network (ANN) introduction, Biological vs Artificial neuron, Simple neuron model, Learning concept (very basic), ANN applications in control	8

Course Outcome (CO)		Bloom's Level
CO1	Describe the basics of control systems along with different types of feedback and its effect. Additionally they will also be able to explain the techniques such as block diagrams reduction, signal	K1

	flow graph and modelling of various physical systems along with modelling of DC servomotor.	
CO2	Explain the concept of state variables for the representation of LTI system.	K2
CO3	Interpret the time domain response analysis for various types of inputs along with the time domain specifications.	K3, K4
CO4	Distinguish the concepts of absolute and relative stability for continuous data systems along with different methods.	K4

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate

TextBook:

1. I.J. Nagrath & M. Gopal, "Control System Engineering", 6th Ed. New Age International Publishers, 2018
2. B.C. Kuo & Farid Golnaraghi, "Automatic Control Systems", 9th Edition, John Wiley India, 2008

Reference Books:

- (Schaums Outlines Series) Joseph J. Distefano III, Allen R. Stubberud, Ivan J. Williams, "Control Systems", 3rd Edition, TMH, Special Indian Edition, 2010.
- A. Anand Kumar, "Control Systems", Second Edition, PHI Learning private limited, 2014.
- William A. Wolovich, "Automatic Control Systems", Oxford University Press, 2011.

CEC 403- ANALOG COMMUNICATION

Pre-requisites: Basic idea of signals, engineering Mathematics

Course Objectives:-

1. Define various fundamental aspects of the communication systems.
2. Understand various modulation & demodulation techniques used in communication systems.
3. Interpret various radio transmitter & receiver circuits and different types of noise in communication systems.
4. Analyze various parameters such as modulation index, channel capacity, transmission efficiency, S/N ratio etc. used in communication systems.

Unit	Topic	Lectures
I.	Introduction Overview of communication system, communication channel, Need for modulation, Review of signals and system, frequency domain representation of signals, principles of amplitude modulation systems- DSB, SSB and VSB modulations.	8
II.	Continuous wave Modulation AM, Angle modulation representation of FM and PM signals, spectral characteristics of angle modulated signals. FM Modulators and Demodulators, FM Broadcasting.	8
III.	Random Processes and Noise Review of probability and random process, Gaussian and white noise characteristics, noise in amplitude modulation systems, noise in frequency modulation systems, pre-emphasis and de-emphasis, threshold effect in angle modulation.	8
IV.	Pulse Modulation Pulse modulation, sampling process, pulse amplitude and pulse code modulation (PCM), differential pulse code modulation. Delta modulation, noise considerations in PCM, time division multiplexing, digital multiplexers.	8
V.	Digital modulation schemes-phase shift keying, frequency shift keying, quadrature amplitude modulation, continuous phase modulation and minimum shift keying.	8

Course Outcome (CO)		Bloom's Level
CO1	Compare different analog modulation schemes for their efficiency and bandwidth.	K1
CO2	Analyze the behavior of a communication system in presence of noise.	K2
CO3	Investigate pulsed modulation system and analyze their system performance.	K3
CO4	Investigate various multiplexing techniques.	K4
CO5	Analyze different digital modulation schemes and compute the bit error performance.	K4

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate

Text Books:-

1. P Ramakrishna Rao., "Communication Systems", Mc Graw Hill Education
2. Haykin S., "Communications Systems", John Wiley and Sons, 2001.
3. Proakis J.G. and Salehi M., "Communication Systems Engineering", "Pearson Education, 2002.

Reference books:-

- Taub H. and Schilling D.L., "Principles of Communication Systems", Tata Mc Graw Hill, 2001.
- Wozencraft J.M. and Jacobs I.M., "Principles of Communication Engineering", John Wiley, 1965.
- B.P.Lathi., "Modern Digital and Analog Communication Systems" fourth edition.
- Barry J.R., Lee E.A. and Messerschmitt D.G., "Digital Communication", Kluwer Academic Publishers, 2004.

CEC 404: ELECTROMAGNETIC FIELD THEORY AND WAVES

Pre-requisite: Engineering Mathematics, Intermediate Physics

Course Objectives:

1. Understand the fundamentals of vector calculus, electrostatics, and magnetostatics.
2. Describe static and dynamic electric and magnetic fields with the help of Maxwell's equations.
3. Analyze the behavior of plane waves in different media.
4. Understand wave phenomena in transmission line.
5. Examine the phenomenon of wave propagation and reflection in different media.

Unit	Topics	Lectures
I.	Vector Calculus: Scalar and vector fields, vector representation of line, surface, Volume integral, gradient, divergence, and curl; divergence and Stokes' theorem; different coordinate systems.	8
II.	Electrostatics and Magnetostatics: Coulomb's law, surface and volume charges, electrostatic potential, Gaussian law and its applications in field determination, Laplace's and Poisson's equation, Ampere's law, Biot-Savart's law, and magnetic flux density, magnetic vector potential, magnetic forces, Faraday's law, boundary conditions at electric and magnetic interfaces.	8
III.	Time Varying Fields and Maxwell's Equations: Displacement current, Maxwell's equation (integral & differential form) - for static, time varying and harmonically varying fields, Poynting theorem and power flow, complex pointing vector, properties of conductors and dielectrics, wave equations for free space and conductors.	8
IV.	Uniform Plane Waves: Uniform plane wave propagation, their transverse nature, reflection by ideal conductor: normal incidence, reflection and transmission with normal incidence at another dielectric, plane wave in lossy dielectric, surface & wave impedance and propagation constant, depth of penetration. Transmission Lines and Smith Chart	8
V.	Application of EMFT with AI Integration: Basics of Electromagnetic Fields (Coulomb's Law, Gauss Law, Maxwell's Equations), Limitations of traditional analytical EMFT methods, Introduction to AI in EMFT, Role of Machine Learning (ML) in electromagnetic modeling, Data-driven vs physics-based modelling. Applications: antenna design, wave propagation, EM simulation	8

Course Outcome (CO)		Bloom's Level
CO1	Apply vector calculus in co-ordinate representation and analyze the static behavior of electric and magnetic fields.	K1 and K3

CO2	Analyze the time varying fields using Maxwell's equations.	K4
CO3	Investigate the characteristics of electromagnetic wave and its propagation in free space.	K3, K4
CO4	Analyze the behavior of electromagnetic wave and its propagation in transmission line.	K4
CO5	Investigate the behavior of ground, space and sky wave propagation.	K4

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate

Text Books:

1. Sadiku, M.N.O, Principles of Electromagnetics, Oxford University Press, 6th edition
2. Hayt, W.H., Engineering Electromagnetic, Tata Mc Graw, 8th edition.
3. John D Krauss, Ronald J Marhefka and Ahmad S. Khan, "Antennas and Wave Propagation", 4th, Tata McGraw Hill.

Reference Books:

- Kraus, J.D., Electromagnetics, Mc Graw-raus, J.D., 5th edition.
- B.N, Basu, Engineering Electromagnetics Essentials, Universities Press.

NPTEL Lectures Link:

1. <https://archive.nptel.ac.in/courses/108/104/108104087/>
1. https://onlinecourses.nptel.ac.in/noc21_ee83/preview

CEC-405: SEMICONDUCTOR FABRICATION TECHNOLOGY

Course Objectives:

1. Understand the fundamentals of semiconductor materials, crystal growth techniques, wafer preparation, and material characterization used in integrated circuit manufacturing
2. Explain the various semiconductor fabrication processes such as oxidation, diffusion, ion implantation, epitaxy, photolithography, etching, and thin-film deposition.
3. Analyze process parameters and their impact on device performance, yield, reliability, and manufacturing efficiency.
4. Understand the fabrication flow of semiconductor devices and integrated circuits, including CMOS process technology and advanced manufacturing techniques.
5. Study cleanroom practices, contamination control, process integration, packaging, testing, and reliability aspects of semiconductor manufacturing.

Unit	Topics	Lectures
I	Introduction To IC Technology: SSI, MSI, LSI, VLSI Integrated Circuits. Crystal Growth and Wafer Preparation: Electronic Grade Silicon, Czochralski Crystal Growth, Silicon Shaping, Processing Considerations. Wafer Cleaning Technology- Basic Concepts, Wet cleaning, Dry cleaning	8
II	Epitaxy: Vapor-Phase Epitaxy, Molecular Beam Epitaxy, Silicon on Insulators, Epitaxial Evaluation. Oxidation: Growth Kinetics, Thin Oxides, Oxidation Techniques and Systems, Oxides Properties.	8
III	Lithography: Optical Lithography, Electron beam lithography, Photomasks, Wet Chemical Etching. Dielectric and Polysilicon Film Deposition: Deposition Processes of Polysilicon, Silicon Dioxide, Silicon Nitride.	8
IV	Diffusion: Models of diffusion in solids, Fick's 1-Dimensional diffusion equation, Diffusion of Impurities in Silicon and Silicon Dioxide, Diffusion Equations, Diffusion Profiles, Diffusion Furnace, Solid, Liquid and Gaseous Sources, Ion-Implantation: Ion-Implantation Technique, Range Theory, Implantation Equipment.	8
V	Metallization: Metallization Application, Metallization Choices, Physical Vapor Deposition, Vacuum Deposition, Sputtering Apparatus. Advanced Topics & Future Trends : AI-driven chip design automation (self-learning tools), Quantum computing & VLSI , 3D ICs and AI applications , Low-power AI chip design, Security challenges in AI chips ,Ethical considerations in AI hardware	8

Course Outcome (CO)		Bloom's Level
CO1	Interpret the basics of crystal growth, wafer preparation and wafer cleaning.	K1, K2
CO2	Evaluate the process of Epitaxy and oxidation.	K2, K4
CO3	Differentiate the lithography, etching and deposition process.	K2, K3

CO4	Analyze the process of diffusion and ion implantation	K1, K4
CO5	Express the basic process involved in metallization and packaging.	K4

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate

Text Books:

- 1.S.M.Sze, "VLSI Technology", Mc Graw Hill Publication, 2nd Edition 2017
- 2.S.K.Gandhi, "VLSI Fabrication Principles", Willy-India Pvt. Ltd, 2008

Reference Books:

- J.D. Plummer, M.D. Deal and Peter B. Griffin, "Silicon VLSI Technology: Fundamentals, Practice and Modeling", Pearson Education Publication, 2009
- Stephen A. Campbell, "Fabrication Engineering at the Micro and Nano scale", Oxford University Press, 2013

CEC-451: MICROPROCESSOR & MICROCONTROLLER LAB

Course Objectives:

1. Understand the architecture, instruction set, and programming environment of microprocessors and microcontrollers through hands-on experimentation.
2. Develop, execute, and debug assembly language programs for arithmetic, logical, data manipulation, and control applications.
3. Implement memory and peripheral interfacing techniques using input/output devices, timers, counters, and interrupt mechanisms.
4. Design and test microcontroller-based systems for real-time monitoring, control, and automation applications.
5. Utilize development boards, simulators, and software tools for program development, hardware interfacing, and system verification.

SUGGESTIVE LIST OF EXPERIMENTS:

1. Write a program using 8085 Microprocessor for Decimal, Hexadecimal addition and subtraction of two Numbers. *(Through Virtual Lab Link)*
2. Write a program using 8085 Microprocessor for addition and subtraction of two BCD numbers. *(Through Virtual Lab Link)*
3. To perform multiplication and division of two 8 bit numbers using 8085. *(Through Virtual Lab Link)*
4. To find the largest and smallest number in an array of data using 8085 instruction set.
5. To write a program using 8086 to arrange an array of data in ascending and descending order. *(Through Virtual Lab Link)*
6. To convert given Hexadecimal number into its equivalent ASCII number and vice versa using 8086 instruction set.
7. To convert given Hexadecimal number into its equivalent BCD number and vice-versa using 8086 instruction set.
8. To interface 8253 programmable interval timer and verify the operation of 8253 in six different modes.
9. To write a program to initiate 8251 and to check the transmission and reception of character.
10. Serial communication between two 8085 through RS-232C port.
11. Write a program of Flashing LED connected to port 1 of the 8051 Micro Controller
12. Write a program to generate 10 kHz square wave using 8051.
13. Write a program to show the use of INT 0 and INT 1 of 8051.
14. Write a program for temperature & to display on intelligent LCD display.

Virtual Lab Link: http://vlabs.iitb.ac.in/vlabs-dev/labs_local/microprocessor/labs/explist.php

Available on: <http://www.vlab.co.in/broad-area-electronics-and-communications>

Course Outcomes:

CO	Course Outcome	Bloom's Level
CO1	Use techniques, skills, modern engineering tools, instrumentation and software/hardware appropriately to list and demonstrate arithmetic and logical operations on 8 bit data using microprocessor 8085.	K4
CO2	Examine 8085&8086 microprocessor and its interfacing with peripheral devices.	K2
CO3	State various conversion techniques using 8085 & 8086 and generate wave forms using 8085.	K2,3
CO4	Implement programming concept of 8051 Microcontroller.	K2
CO5	Design concepts to Interface peripheral devices with Microcontrollers o as to design Microcontroller based projects.	K3

K1- Remember, K2- Understand, K3- Apply, K4- Analyze, K5- Evaluate

CEC-452: CONTROL SYSTEM LAB

Course Objectives:

1. Understand and verify the fundamental concepts of control systems through simulation and experimental analysis of dynamic systems.
2. Develop mathematical models of control systems and analyze their behavior using transfer functions, block diagrams, and state-space representations.
3. Study and evaluate the time-domain response characteristics of control systems, including rise time, settling time, peak overshoot, and steady-state error.
4. Analyze the stability of control systems using techniques such as Root Locus, Bode Plot, Nyquist Plot, and Routh–Hurwitz Criterion through software-based experiments.
5. Design and implement controllers and compensators to improve system performance and meet specified design requirements.

SUGGESTIVE LIST OF EXPERIMENTS:

1. Introduction to MATLAB Control System Toolbox.
2. Determine transpose, inverse values of given matrix.
3. Plot the pole-zero configuration in s-plane for the given transfer function.
4. Determine the transfer function for given closed loop system in block diagram representation.
5. Create the state space model of a linear continuous system.
6. Determine the State Space representations of the given transfer function.
7. Determine the time response of the given system subjected to any arbitrary input.
8. Plot unit step response of given transfer function and find delay time, rise time, peak time, peak overshoot and settling time.
9. Determine the steady state errors of a given transfer function.
10. Plot root locus of given transfer function, locate closed loop poles for different values of k.
11. Plot bode plot of given transfer function. Also determine gain and phase margins.
12. Plot Nyquist plot for given transfer function. Also determine the relative stability by measuring gain and phase margin.

Course Outcomes:

CO	Course Outcome	Bloom's Level
CO1	Classify different tools in MATLAB along with the basic matrix operations used in MATLAB.	K2
CO2	Evaluate the poles and zeros on s-plane along with transfer function of a given system.	K2
CO3	Construct state space model of a linear continuous system.	K3
CO4	Evaluate the various specifications of time domain response of a given system.	K4
CO5	Appraise the steady state error of a given transfer function.	K4
CO6	Examine the relative stability of a given transfer function using various methods such as root locus, Bode plot and Nyquist plot.	K4

K1- Remember, K2- Understand, K3- Apply, K4- Analyze, K5- Evaluate

CEC-453: COMMUNICATION LAB-I

Course Objectives:

1. Understand the operating principles of analog communication systems through the implementation and analysis of amplitude modulation (AM), frequency modulation (FM), and their corresponding demodulation techniques.
2. Perform experiments on pulse modulation techniques such as PAM, PWM, PPM, PCM, and Delta Modulation, and analyze their generation, transmission, and recovery.
3. Develop practical skills in implementing and analyzing digital modulation techniques, including FSK and PSK, using communication trainer kits and simulation tools.
4. Measure, observe, and interpret communication signals using laboratory instruments such as Digital Storage Oscilloscopes (DSO/CRO), function generators, and spectrum analyzers where applicable.
5. Compare the performance of various analog and digital communication techniques by evaluating parameters such as modulation index, bandwidth, signal quality, and noise immunity.

SUGGESTIVE LIST OF EXPERIMENTS:

1. Study of DSB-SC (Double Sideband Suppressed Carrier) Amplitude Modulation and determination of modulation index and sideband power.
2. Study of Amplitude Demodulation using a linear diode detector.
3. Study of Frequency Modulation (FM) and determination of modulation index.
4. Study of Pulse Amplitude Modulation (PAM) – Sampling and reconstruction.
5. Study of Pulse Width Modulation (PWM) generation and demodulation.
6. Study of Pulse Position Modulation (PPM) generation and demodulation.
7. Study of Pulse Code Modulation (PCM) generation and detection.
8. Study of Delta Modulation (DM) and demodulation.
9. Generation and detection of Frequency Shift Keying (FSK).
10. Generation and detection of Phase Shift Keying (PSK/BPSK).

Course Outcomes:

CO	Course Outcome	Bloom's Level
CO1	Understand the operating principles of analog communication systems through the implementation and analysis of amplitude modulation (AM), frequency modulation (FM), and their corresponding demodulation techniques.	K2
CO2	Perform experiments on pulse modulation techniques such as PAM, PWM, PPM, PCM, and Delta Modulation, and analyze their generation, transmission, and recovery.	K2
CO3	Develop practical skills in implementing and analyzing digital modulation techniques, including FSK and PSK, using communication trainer kits and simulation tools.	K3
CO4	Measure, observe, and interpret communication signals using laboratory instruments such as Digital Storage Oscilloscopes (DSO/CRO), function generators, and spectrum analyzers where applicable.	K4
CO5	Compare the performance of various analog and digital communication techniques by evaluating parameters such as modulation index, bandwidth, signal quality, and noise immunity.	K4

K1- Remember, K2- Understand, K3- Apply, K4- Analyze, K5- Evaluate

CEC 456: Mini Project (01 Credit)

Course Objectives:

The student is required to undertake a short project based on the syllabus. The objective is to provide hands-on training in the practical aspects of the curriculum and to enhance their research and analytical skills.

CVA 402: UNIVERSAL HUMAN VALUES

Course Objectives:

The objectives of this course are:

1. To help students understand the importance of human values in personal and professional life
2. To develop a holistic perspective towards life, society, and nature
3. To enable students to differentiate between values and skills, and understand their role in harmony
4. To promote ethical and responsible behavior in society
5. To encourage self-exploration and continuous self-improvement

Syllabus

Content	Contact Hours
Unit-1:Introduction to Human Values	8
Need, Basic Guidelines, Content and Process for Value Education; Self-Exploration: What is it? Its content and process; ‘Natural Acceptance’ and Experiential Validation; Continuous Happiness and Prosperity; Right understanding, Relationship and Physical Facility; Understanding the Human Being as Co-existence of Self (I) and Body.	
Unit-2:Understanding Harmony in the Human Being	8
Harmony in the Human Being: Understanding the needs of Self (I) and Body; Understanding the Body as an Instrument of ‘I’; Harmony in ‘I’ (Self): Thoughts, Behavior, Work; Harmony of Self with Body; Programs to ensure self-regulation and health.	
Unit-3:Understanding Harmony in the Family and Society	8
Harmony in Family: Trust and Respect as the foundational values; Other feelings: Affection, Care, Guidance, Reverence, Glory, Gratitude, Love; Harmony in Society: Understanding values in human relationships; Justice in human-to-human relationships; Role of family in society.	
Unit-4:Understanding Harmony in Nature and Existence	8
Harmony in Nature: Interconnectedness and mutual fulfillment among four orders of nature (Material, Plant, Animal, Human); Understanding existence as co-existence; Environmental sustainability; Role of human beings in maintaining harmony with nature.	
Unit-5:Implications of Human Values in Professional Ethics	8
Professional Ethics: Natural acceptance of human values; Ethical human conduct; Competence in professional ethics; Holistic technologies, production systems and management models; Case studies on ethical practices; Social responsibility and sustainable development.	

Course Outcome:

At the end of course, the student will be able to

Course Outcome (CO)		Bloom's Level
CO1	Understand the significance of human values and self-exploration	K2
CO2	Analyze harmony within self and between self and body	K4
CO3	Apply values to improve relationships in family and society	K3
CO4	Understand harmony in nature and environmental sustainability	K2
CO5	Apply ethical principles in professional and social life	K4

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate

Textbooks

1. A Foundation Course in Human Values and Professional Ethics – R.R. Gaur, R. Sangal, G.P. Bagaria

Reference Books:

- Human Values – A.N. Tripathi
- Professional Ethics and Human Values – M. Govindarajan, S. Natarajan, V.S. Senthil Kumar